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A3 - scholarship on reserved topics

Funded by: University of Trento - Department of Civil, Environmental and Mechanical Engineering within the project “Seismic Risk” CUP no. C65E24000330003 – PAT

Title: Next Generation of Exposure and Seismic Risk Maps for the Trentino region

This research project aims to develop a prototype for high-resolution seismic risk maps at the regional scale, integrating microzonation studies with advanced vulnerability and exposure assessments. The project develops a comprehensive framework that integrates state-of-the-art predictive models of seismic hazard, structural vulnerability including soil-structure interaction, and exposure data for improved risk estimations and more efficient mitigation strategies across the Trentino region. The research outcomes are structured into three key contributions:

1. Local-Scale Seismic Hazard Assessment and Microzonation Integration

- Development of a methodological framework to incorporate microzonation studies into seismic hazard models. The integration will be fully probabilistic.
- Comparison of national-scale hazard assessments with local microzonation data, enhancing accuracy and resolution in risk estimation.

2. Seismic Vulnerability and Exposure Modeling of the Built Environment

- State-of-the-art vulnerability models based on advanced surrogate models, and integration of soil-structure interaction according to computationally efficient, thermodynamic-based macroelement representations.
- Integration of national building datasets with Global Earthquake Modelling taxonomies to map structural vulnerabilities.
- Identification of high-risk building typologies and strategic infrastructure, enabling targeted risk mitigation.

3. Integrated Seismic Risk Mapping: A Case Study in Trentino

- Development of a GIS-based seismic risk assessment model that combines hazard, vulnerability, and exposure data.

- Validation through application to the Alto Garda region, demonstrating adaptability for broader regional and national applications.
- Through the systematic comparison with conventional microzonation models, identification of urban aggregates in which soil-structure interaction can have negligible, favourable or detrimental effects on seismic risk.

This project is a pioneering effort in Italy and at the forefront of international research, integrating high-resolution microzonation with exposure analysis to develop a replicable and scalable seismic risk assessment methodology. The findings enhance seismic risk management and align with European initiatives such as ESRM20 (European Seismic Risk Model).

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A4 - scholarship on reserved topics

Funded by: University of Trento – [Eurac Research](#)

Title: Climate Scenarios for Future Water Availability in Alpine Regions

Climate change and human activities are reshaping water availability and demand in the Alps. Changes in precipitation and temperature patterns, higher water needs for mountain ecosystems in response to warming and increasing water demands from socio-economic activities are expected to threaten water security in the next future. In the upper Adige River catchment (Eastern Italian Alps), reduced snowfall and glacier melt have already shifted the streamflow regime, and impacts of the summer streamflow decrease have already been registered in downstream areas. Recent events, such as the 2021–2022 drought, generated tensions and disputes for sharing the available water from upstream to downstream. This situation highlighted the need for improving the understanding and modelling of climate change effects on current and future water availability.

This PhD project aims to comprehensively address the challenges related to the modelling of current and future water availability in the Alps by improving and integrating information on climate change, snow dynamics, and evapotranspiration processes into hydrological simulations with a focus on the upper Adige catchment (Trentino-South Tyrol). The overall research question behind is: “How do changes in climate drivers, cryosphere and evapotranspiration demand influence the hydrological regime of mountain catchments, from an upstream-to-downstream perspective?”

Specific tasks will include: 1) evaluation of climate data for hydrological modelling, analysis of climatic drivers and development of meaningful climate scenarios for a critical assessment of future water resources and impacts of water scarcity conditions; 2) assessment of evapotranspiration contributions to water availability by integrating climatic data and land-use scenarios and developing and testing specific modelling modules; 3) study of spatio-temporal snow dynamics, including snowmelt processes, for improving its representation in hydrological models; 4) the inclusion of such processes in a multi model framework for current and future scenarios.

Expected research outputs will include at least three peer-reviewed scientific publications in high-impact journals covering the specific objectives listed above.

This PhD project will be partly supported by the NextWater_ST project, funded by the Province of Bolzano and led by Eurac Research (a period of the doctoral activities is required to be spent at Eurac premises in Bolzano). The project aims to enhance the understanding of the dynamics between water availability and multi-sectoral water demand in South Tyrol under future changes to support water management.

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